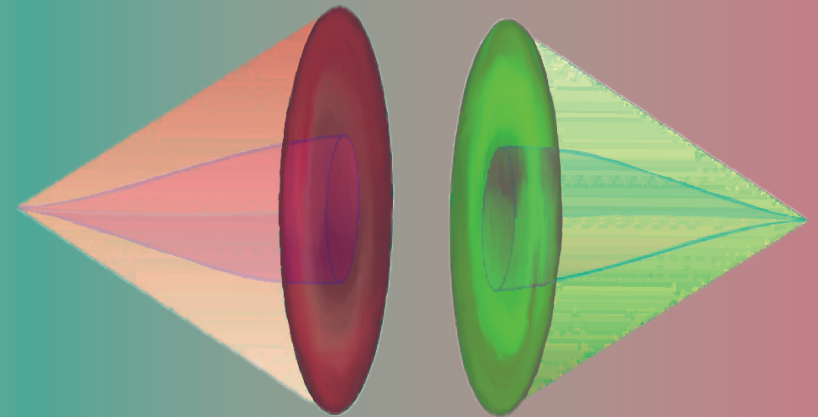
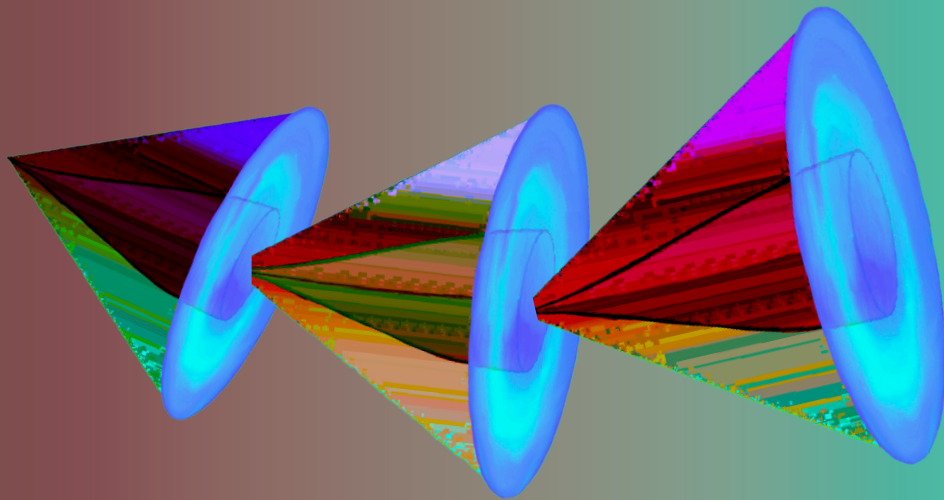


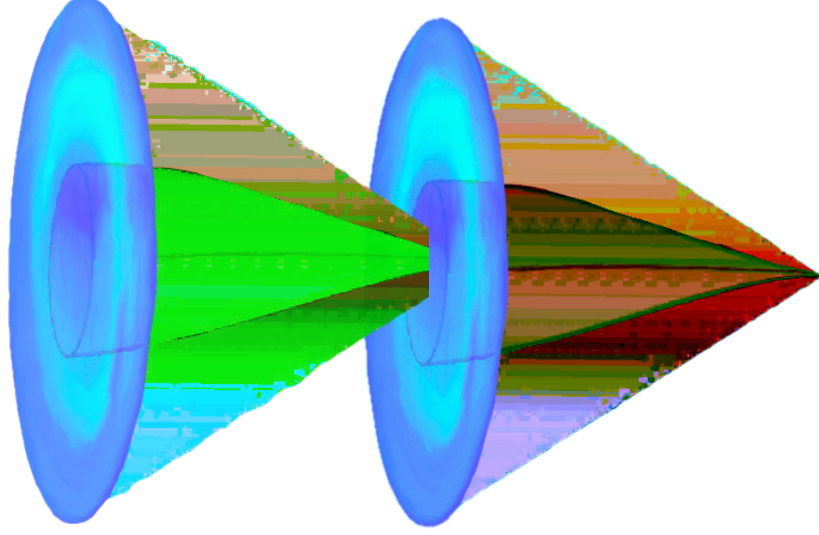
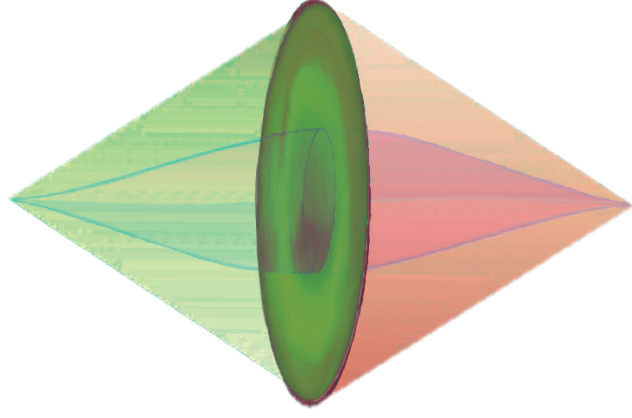
Frictional Technologies:

Speculative Methods for
Critique and Collaboration



Thank you for working on this exercise with us!

This is version 1.0 of this booklet. If you have comments or critiques, we would love to hear them. You can view this work online and send us feedback or completed booklets at:



Finally, can you think of any people or groups who research or work towards systems like the ones you drew?

Do you have any friends, peers, or mentors who might want to join you?

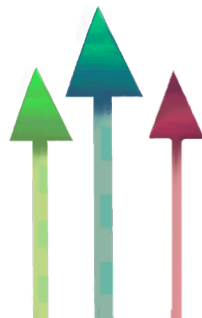
List some below:

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This exercise will help you understand and re-imagine everyday technologies using speculative thinking.

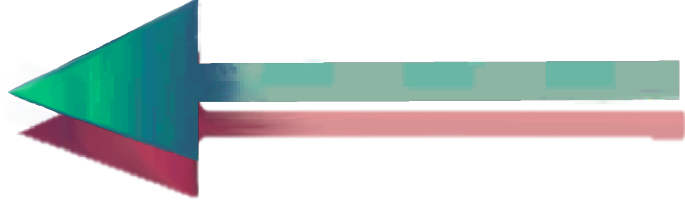
It includes three parts. First, you'll find your initial **inputs**, then your **critique**, and finally, your actionable **desire**.

Step-by-step, you will build speculative knowledge as you engage with your own experience, emotion, and perspective.



PART 1:

CHOOSING YOUR INPUTS

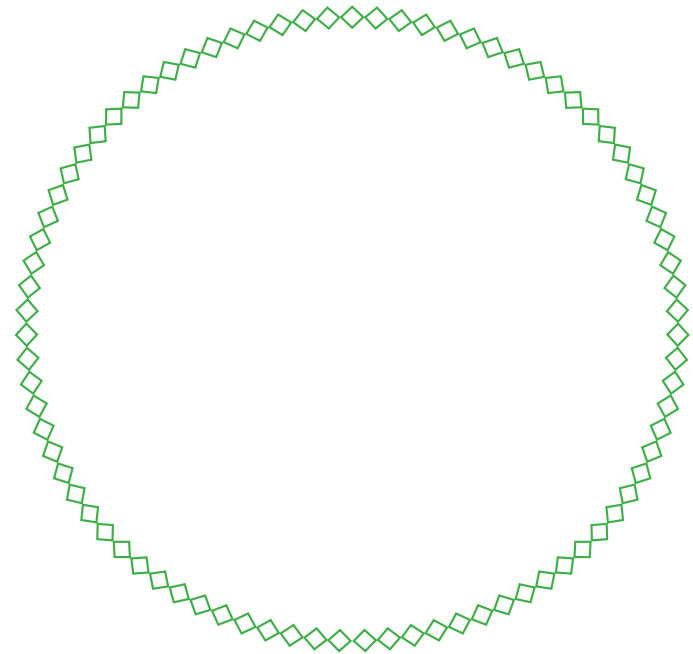


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Second, what social systems, computing paradigms, or cultural knowledge sets do your new systems rely on?

Are these systems, paradigms, or knowledge sets that you're familiar with? If you wanted to learn more about them, where would you start?

1. Select a technology or system that you use regularly, but find intimidating or overwhelming. Write it in the circle below.



If you can't think of any, here are some examples: a coding environment; a file management system; your preferred email platform; a go-to device (like a phone, computer, or television); or an interactive chatbot. You could also choose a non-digital system, such as public transit or your doctor's office.

On the next page, use words, a diagram, or a drawing to describe your memory.

2. Describe a memory of a frustrating interaction you've had with this system.

Make sure to answer:

Which system components were particularly troublesome?

What were you trying to do that became difficult, overwhelming, or irritating?

Did you experience multiple points of tension?

Do you regularly have the same issues with this system?

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4.

In the very last step of this exercise, you will respond to three sets of questions that point you towards future research and collaboration.

First, consider:

What sorts of materials does your new system require?

Who grows, creates, researches, or thinks about those materials?



3. Use this memory to name a "**friction**."

A "**friction**" is a frustrating interaction between you and a system.

It is what happens when your needs or wants don't match some pre-designed component of a system.

To name your "**friction**," comb through the memory on page 5.

First: which component of your larger system were you struggling with especially?

- **Component:**

Second: in a sentence, summarize what made your interaction frictional.

- **Friction:**

4.

Now that you've named your
friction, let's switch gears.

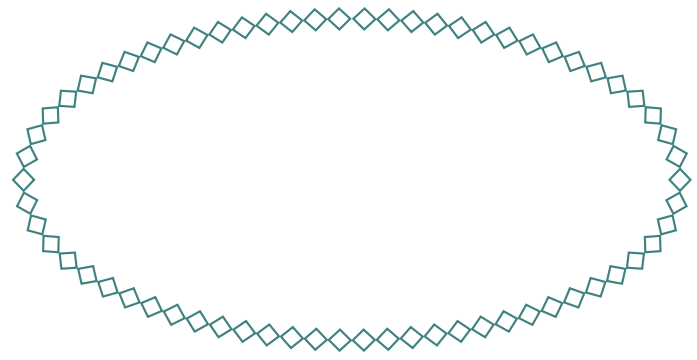
On this page, write a few notes
about something exciting, curious,
or hopeful you learned in this past
week.

→ This might come from an interesting conversation ←
with a friend or peer; a book or article; a lecture, workshop, or panel;
or your own observations of the world around you.

5. Lastly, choose a **value** based on your recent lesson.

Your value should explain what struck you about this new information. Your choice can be as analytical or intuitive as you'd like.

Write the value in the area below.



Sample values
you might
choose from:

Variability	Autonomy	Efficiency	Education	Connection	Balance	Responsiveness
Approachability	Challenge	Collaboration	Playfulness	Productivity	Restoration	Equality

3. Imagine: another person comes along and resolves that **barrier**.

To do that, they use your selected **value** to apply some (currently) impossible technical or material process. Suspend disbelief: picture the system now meeting the conditions of your countervalence.

On the next pages, draw two visions of what interactions with your previously frictional system might look like now.

Don't worry about artistic quality: this is just to get ideas on paper.

2. What **barriers** stand between your countervalance being possible and real life?

These might be the system's own technical limitations, gaps in your skills or knowledge, material parameters, or systemic gaps.

Choose one barrier to focus on.

- Your barrier:

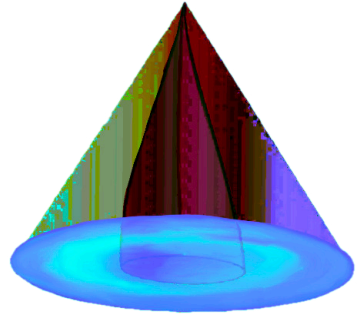
Rewrite each input below. This will be convenient later in the exercise, when you refer back to this list.

- System:
- Component:
- Friction:
- Value:

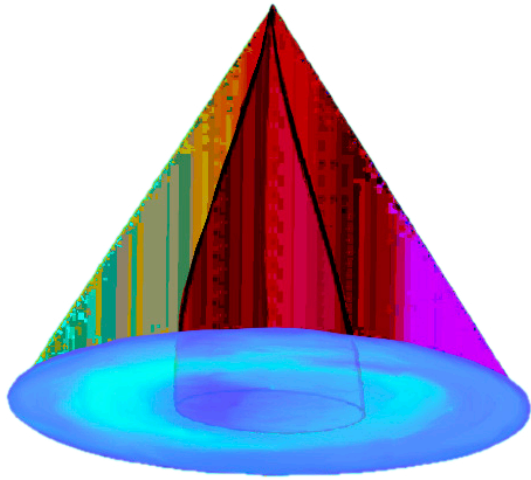


PART 2: FINDING YOUR CRITIQUE

$$\left(\left(\frac{\text{friction}}{\text{system}} \right) = \text{design goal} \right) \leftrightarrow \text{why} = \text{motivation} \times \text{tension} \times \text{value} = \text{critique}$$



- 1. First, flip your critique on its head.
- Create a **countervalue**, or a statement that exactly opposes your critique.
- This will shift us from your initial system-component to a new, re-imagined one.
- Countervalue:**



1. First, let's figure out why the friction is happening.

Within the larger system, what role does your component play?

How is this component supposed to help you interact with the system?

How might a designer have expected you to interact?

Use these questions to imagine one **design goal** (an experiential, behavioral, or functional target) that influences your component.

PART 3:

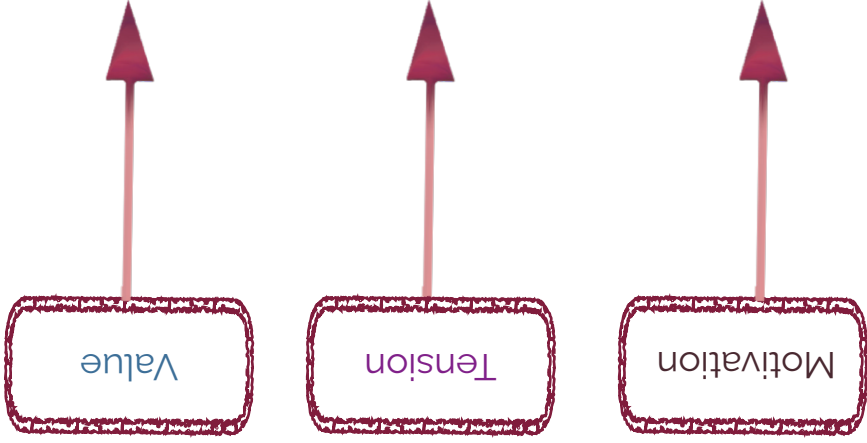
SPECULATIONS AND COLLABORATIONS

- Design goal:

2. Let's dig deeper. What is the motivation behind that **goal**?
Repeatedly ask "why" the goal might have been chosen until you identify a root purpose.*
If you get stuck, ask yourself: why would a designer think that you, as a user, might find this feature useful? How could it benefit you?
How could it benefit the manufacturers? The company? The funders?

As you move through "whys," you will shift focus from the component to the broader system.

This is your **critique**.



4. Lastly, let's tie together our work from the last few pages.

Look at your **motivation**, your **tension**, and the **value** you identified on page X. Then, write a few words or a sentence that condense your thinking into a **critique**.

To write this **critique**, you might consider:

How do my own personal or political **tensions** clash with the design's underlying motivation?

How does that motivation limit the **value** I chose?

Use the next page to write your critique.

Goal 1: _____

why?

Goal 2: _____

why?

Goal 3: _____

why?

Goal 4: _____

why?

Goal 5: _____

Tension



14.

3. Now, let's turn to **you**.
Is anything about you specifically
in tension with this **motivation**?
Do your lifestyle, location, type of
work, learning methods, politics,
capacities, or interests clash with it
in any way?
Respond to this question on the
next page.

Tension

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15.

